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1 import pandas as pd
2 import numpy as np
3 from scipy.stats import norm
4
5 # Transpose ข้อมูล
6 df = pd.read_excel("/content/Book5.xlsx")
7 df_T = df.T
8 df_T.columns = df_T.iloc[0]
9 df_T = df_T.iloc[1:].reset_index(drop=True)
10
11 # กรองเฉพาะตัวแปรที่เป็นตัวเลข
12 df_numeric = df_T.drop(columns=["!Sample_characteristics_ch1"], errors="ignore")
13 df_numeric = df_numeric.apply(pd.to_numeric, errors="coerce").fillna(0)
14
15 # กรองเฉพาะตัวแปรค่า CV อยู่ในช่วง [0.25,0.5]
16 def filter_columns_by_cv_only(df_numeric, cv_range=(0.25, 0.5)):
17     mean_vals = df_numeric.mean(axis=0)
18     std_vals = df_numeric.std(axis=0, ddof=1)
19     cv_vals = std_vals / mean_vals.replace(0, np.nan)
20     cond_cv = (cv_vals >= cv_range[0]) & (cv_vals <= cv_range[1])
21     kept_cols = df_numeric.columns[cond_cv]
22     return kept_cols
23
24 kept_cols = filter_columns_by_cv_only(df_numeric)
25 print(f"จำนวนตัวแปรที่เหลือหลังการกรอง CV (0.25-0.5): {len(kept_cols)}")
26 cols_to_keep = ["!Sample_characteristics_ch1"] + list(kept_cols)
27 df_T = df_T[cols_to_keep]
28
29 # แบ่งกลุ่มข้อมูล
30 def clean_and_select(df_T, keywords):
31     df_sub = df_T[df_T["!Sample_characteristics_ch1"].isin(keywords)].copy()
32     df_sub = df_sub.drop(columns=["!Sample_characteristics_ch1"])
33     df_sub = df_sub.apply(pd.to_numeric, errors="coerce").fillna(0)
34     return df_sub
35 df_AT = clean_and_select(df_T, [
36     "Histopathological diagnostic: astrocytoma, grade 3",
37     "Histopathological diagnostic: astrocytoma, grade 2"
38 ])
39 df_GBT = clean_and_select(df_T, [
40     "Histopathological diagnostic: glioblastoma, grade 4",
41     "Histopathological diagnostic:"
42 ])
43 df_ODG = clean_and_select(df_T, [
44     "Histopathological diagnostic: oligodendroglioma, grade 2",
45     "Histopathological diagnostic: oligodendroglioma, grade 3"
46 ])
47 df_NT = clean_and_select(df_T, [
48     "Histopathological diagnostic: non-tumor"
49 ])
50
51 # ตรวจสอบขนาดของแต่ละกลุ่ม
52 print("\nShapes หลังการกรอง:")
53 for name, df_in zip(["AT", "GBT", "ODG", "NT"], [df_AT, df_GBT, df_ODG, df_NT]):
54     print(name, df_.shape)
55
56 # แปลงเป็น NumPy
57 data_AT, data_GBT, data_ODG, data_NT = map(lambda x: x.to_numpy(), [df_AT,
58     df_GBT, df_ODG, df_NT])
59 data_groups = [data_AT, data_GBT, data_ODG, data_NT]
60
61 # คำนวณสถิติทดสอบ T3
62 EPS = 1e-12
63 def _Bi_within(X):
64     n = X.shape[0]
65     if n < 2:
66         return 0.0
67     G = X @ X.T
68     np.fill_diagonal(G, 0.0)
69     return np.sum(G * G) / (n * (n - 1))
70
71 def _Bij_cross(X, Y):
72     H = X @ Y.T
73     return np.sum(H * H) / (X.shape[0] * Y.shape[0])
74
75 def ahmad_Tg_test(data_groups, alpha=0.05):

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75     g = len(data_groups)
76     n_i = [grp.shape[0] for grp in data_groups]
77     p = data_groups[0].shape[1]
78
79     B_within = [_Bi_within(grp) for grp in data_groups]
80     B_pairs = {}
81     for i in range(g):
82         for j in range(i + 1, g):
83             B_pairs[(i, j)] = _Bij_cross(data_groups[i], data_groups[j])
84
85     sum_Bi = np.sum(B_within)
86     sum_Bij = np.sum(list(B_pairs.values()))
87     Tg = ((g - 1) * sum_Bi - 2.0 * sum_Bij) / (p * p)
88
89     numerator_C3, P_star = 0.0, 0.0
90     for i in range(g):
91         for j in range(g):
92             if i == j:
93                 continue
94             ni, nj = n_i[i], n_i[j]
95             key = (i, j) if i < j else (j, i)
96             Bij = B_pairs[key]
97             numerator_C3 += ni * nj * Bij
98             P_star += ni * nj
99     C3 = numerator_C3 / max(P_star, EPS)
100
101     term1 = (g - 1)**2 * np.sum([1.0 / (ni ** 2) for ni in n_i])
102     term2 = np.sum([2.0 / (n_i[i] * n_i[j]) for i in range(g) for j in range(i
+ 1, g)])
103     A = term1 + term2
104     sigma_hat = 2.0 * np.sqrt(max(A, 0.0)) * (C3 / (p * p))
105     sigma_hat = max(sigma_hat, EPS)
106     stat = Tg / sigma_hat
107     p_value = 1 - norm.cdf(stat)
108     return stat, p_value, Tg, sigma_hat
109
110 # เรียกใช้ฟังก์ชันของสถิติทดสอบ T3
111 stat, p_value, Tg, sigma_Tg0 = ahmad_Tg_test(data_groups)
112 print("\n===== ผลการทดสอบ =====")
113 print(f"Statistic Test (T3) = {stat:.6f}")
114 print(f"p-value           = {p_value:.6f}")

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จำนวนตัวแปรที่เหลือหลังการกรอง CV (0.25-0.5): 24785

Shapes หลังการกรอง:

AT (26, 24785)
 GBT (81, 24785)
 ODG (50, 24785)
 NT (23, 24785)

===== ผลการทดสอบ =====

Statistic Test (T3) = 2.046850
 p-value = 0.020336